Ex. 18 - Anchoring

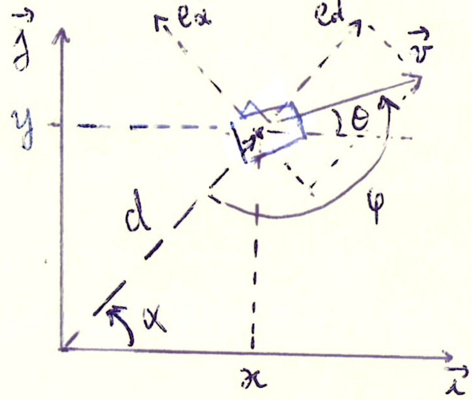
1) $\begin{cases} \dot{x} = \cos\theta \\ \dot{y} = \sin\theta \\ \dot{\theta} = u \end{cases}$ with $X = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix}$

We could define a point of equilibrium if $\exists X$ with $\dot{X} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$

But we have the speed robot $v = \sqrt{\dot{x}^2 + \dot{y}^2} = \sqrt{\cos^2\theta + \sin^2\theta} = 1 \neq 0$

The speed is never null, so there are no point of equilibrium.

2)



Here, we can consider the polar reference with $\vec{e_d}, \vec{e_\alpha}$

The speed can be deduce by:

$\vec{v} = v_d \vec{e_d} + v_\alpha \vec{e_\alpha}$

$= v \cdot \sin(\varphi - \frac{\pi}{2}) \vec{e_d} - v \cdot \cos(\varphi - \frac{\pi}{2}) \vec{e_\alpha}$

but $v = 1$

so $\vec{v} = -\cos\varphi \, \vec{e_d} - \sin\varphi \, \vec{e_\alpha}$

But we also have $\vec{v} = \frac{d}{dt}(\vec{d}) = \frac{d}{dt}(d\vec{e_d}) = \dot{d}\,\vec{e_d} + d \cdot \frac{d}{dt}(\vec{e_d})$

Knowing $\vec{e_d} = \begin{pmatrix} \cos\alpha \\ \sin\alpha \end{pmatrix}_{ij}$ and $\vec{e_\alpha} = \begin{pmatrix} -\sin\alpha \\ +\cos\alpha \end{pmatrix}_{ij}$, we deduce $\frac{d}{dt}(\vec{e_d}) = +\dot{\alpha}\,\vec{e_\alpha}$

So $\begin{cases} \vec{v} = -\cos\varphi\,\vec{e_d} - \sin\varphi\,\vec{e_\alpha} \\ \vec{v} = \dot{d}\,\vec{e_d} + \dot{\alpha} d\,\vec{e_\alpha} \end{cases} \Rightarrow \begin{cases} \dot{d} = -\cos\varphi \\ \dot{\alpha} = -\frac{\sin\varphi}{d} \end{cases}$

Knowing $\varphi - \theta + \alpha = \pi$ from the graphical representation, we deduce

$\dot{\varphi} = \dot{\theta} - \dot{\alpha} = u + \frac{\sin\varphi}{d}$

The final state equation :

$$\boxed{\begin{cases} \dot{\alpha} = -\frac{\sin\varphi}{d} \\ \dot{d} = -\cos\varphi \\ \dot{\varphi} = \frac{\sin\varphi}{d} + u \end{cases}}$$

3) we have $\dot{\varphi} = -\dot{\alpha} + u$ and $\dot{\alpha} = -\cos\varphi$

So if we have $\dot{\varphi}$, we can deduce $\dot{\alpha}$ and so we don't need 3 variables in the state equation to control the vehicle.

4) The goal of $u = \begin{cases} +1 & \text{if } \cos\varphi \leq \frac{1}{\sqrt{2}} \\ -\sin\varphi & \text{otherwise} \end{cases}$ is to keep the boat turning around a virtual point.

φ can represent the alignment of the robot with this virtual point.

Depending on the size and signe of φ, which can be deduced from $\sin\varphi$ or $\cos\varphi$.

With $-\sin\varphi$, the robot will move forward with some left or right moves.

If it reach a specific threshold, it will turn left and will keep the same cap.